# Muhammad Ayaan Afzal

*Iowa State University Mechanical Engineering (Honors Junior)*

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## Profile

Driven mechanical engineering student seeking a summer 2026 co-op/internship in engineering design, manufacturing, or quality assurance.

## Work Experience

**Undergraduate Teaching Assistant - EM 3240 Mechanics of Materials** 01/2026 – Present | Ames, IA, US
*Iowa State University - College of Engineering*
- Support instruction for a class of 120 undergraduate students in Mechanics of Materials.
- Clarify concepts including plane stress, plane strain, stress–strain relationships, material behavior, stress/deformation analysis, failure theories, and buckling.
- Assist students with in-class problem solving and applying concepts to axial, torsional, flexural, and combined loadings.

**Undergraduate Teaching Assistant - ME 1700** 01/2026 – Present | Ames, IA, US
*Iowa State University - College of Engineering*
- Support 30 undergraduate students in engineering graphics, CAD modeling, and introductory design methodologies
- Assist with multiview projections, dimensioning standards, and 3D geometric visualization
- Evaluate and grade technical drawings and design assignments for accuracy, tolerancing, and clarity of engineering communication.

**Undergraduate Teacher Assistant — CE 2740 Engineering Statics** 08/2025 – Present | Ames, IA, US
*Iowa State University - College of Engineering*
- Conduct weekly recitation sessions for ~15 students on engineering statics, effectively communicating complex technical skills to peers.
- Manage high-volume grading of homework and exams for a ~120-student section, deciphering technical content to identify and resolve errors.
- Adapt teaching approaches to meet diverse student needs and collaborate with instructors to ensure consistent grading standards.

**Undergraduate Research Assistant** 01/2025 – Present | Ames, IA, US
*Mechanical Engineering Department - STAMP Group (Prof. Tuhin Mukherjee)*
- Optimized Wire Arc Additive Manufacturing (WAAM) parameters by simulating droplet transfer and weld pool behavior, assessing longitudinal and transverse oscillation impacts on deposition quality.
- Enhanced Fortran-based simulation code to model oscillatory motion and implement bulge formation, improving alignment with experiments.
- Calibrated simulations using thermal and velocity data to closely match experimental conditions, supporting process optimization and defect reduction.
- Conducted parametric studies and used TECPLOT 360 to analyze temperature fields and solidification behavior.

**Product Development Engineering Intern** 12/2025 – 01/2026 | Karachi, Pakistan
*Asia Gas & Electrical Appliances*
- Reverse engineered a compact thermal appliance; recreated CAD model, BOM, and manufacturing documentation from original product.
- Implemented cost-down redesign using DFM/DFA, part standardization, and material/component substitutions.
- Optimized assembly for higher thermal efficiency, improved durability, and reduced shipping weight.
- Developed multi-voltage product variants for EU/US export and supported prototype builds through production release.

## Projects

**Human-Powered Tiller Design Project (ME 2700)** 08/2025 – 12/2025
- Led mechanical design and material selection using SolidWorks, translating market research into engineering constraints for a human-powered tiller.
- Performed DFM/DFA analysis and managed the Bill of Materials (BOM) to optimize part count and strictly adhere to a $150 budget.
- Fabricated a fully functional prototype using industrial machinery (mills, lathes) to validate the design and manufacturing feasibility.

**Makala Soprano Ukulele (Model MK-S)** 07/2025 – 08/2025
- Reverse engineered a Makala soprano ukulele, capturing precise dimensions and geometric details through detailed measurements.
- Designed a high-fidelity 3D CAD model in SolidWorks, incorporating complex surface features and assemblies.
- Produced a high-accuracy physical replica using 3D printing, ensuring precise geometry and visual detail.

**Tire Rim Design for Solar Car — PRISUM Team** 06/2025 – 07/2025
- Designed and 3D printed a custom lightweight tire rim optimized for structural strength and cost-efficiency.
- Applied engineering principles to enhance vehicle dynamics and performance under real-world conditions.
- Integrated aesthetic and functional design to meet project requirements for the solar car team.

**Piston Engine – 4-Cylinder Assembly** 06/2025 – 06/2025
- Modeled a complete 4-cylinder piston engine assembly from scratch, including key components such as crankshaft and pistons.
- Applied precise 3D modeling techniques and mechanical mates to accurately simulate realistic motion and component interaction.
- Developed proficiency in assembly design and constraint management, enhancing understanding of internal combustion engine mechanics.

## Education

**Bachelor of Science in Mechanical Engineering (Honors)** 05/2028 | Ames, IA, U.S
**GPA:** 3.86/4.0

## Skills

SOLIDWORKS | Tecplot 360 | Computer-Aided Design (CAD) | Python | Fortran | MATLAB | MS Excel

## Honors and Awards

**First Place – American Welding Society (AWS) Poster Competition (Undergraduate, B.S. Engineering)**
Achieved 1st place out of 25 student entries and secured a $750 monetary award.